

Answers:

1) a) parallel b) Parallel c) Perpendicular d) Parallel

3) a) parallel b) perpendicular d) perpendicular e) parallel

4) a) $m = \frac{3}{5}$ b) $m = -1$ d) $m = \frac{-4}{3}$ e) $m = 0$

5) a) $m = \frac{-5}{3}$ b) $m = 1$ d) $m = \frac{3}{4}$ e) undefined

6) Any two lines with a slope of $\frac{1}{2}$. Examples: $y = \frac{1}{2}x + 99$ and $y = \frac{1}{2}x + 87$

7) Any two lines with a slope of $\frac{1}{4}$. Examples: $y = \frac{1}{4}x + 71$ and $y = \frac{1}{4}x + 87$

8) b) It appears vertex B might be a 90 degree angle c) $m_{AB} = 3, m_{BC} = \frac{-1}{3}, m_{AC} = 1$

d) $m_{AB} \times m_{BC} = -1$, therefore AB and BC are perpendicular

9) a) Not a right triangle b) PQ and PR form a right angle